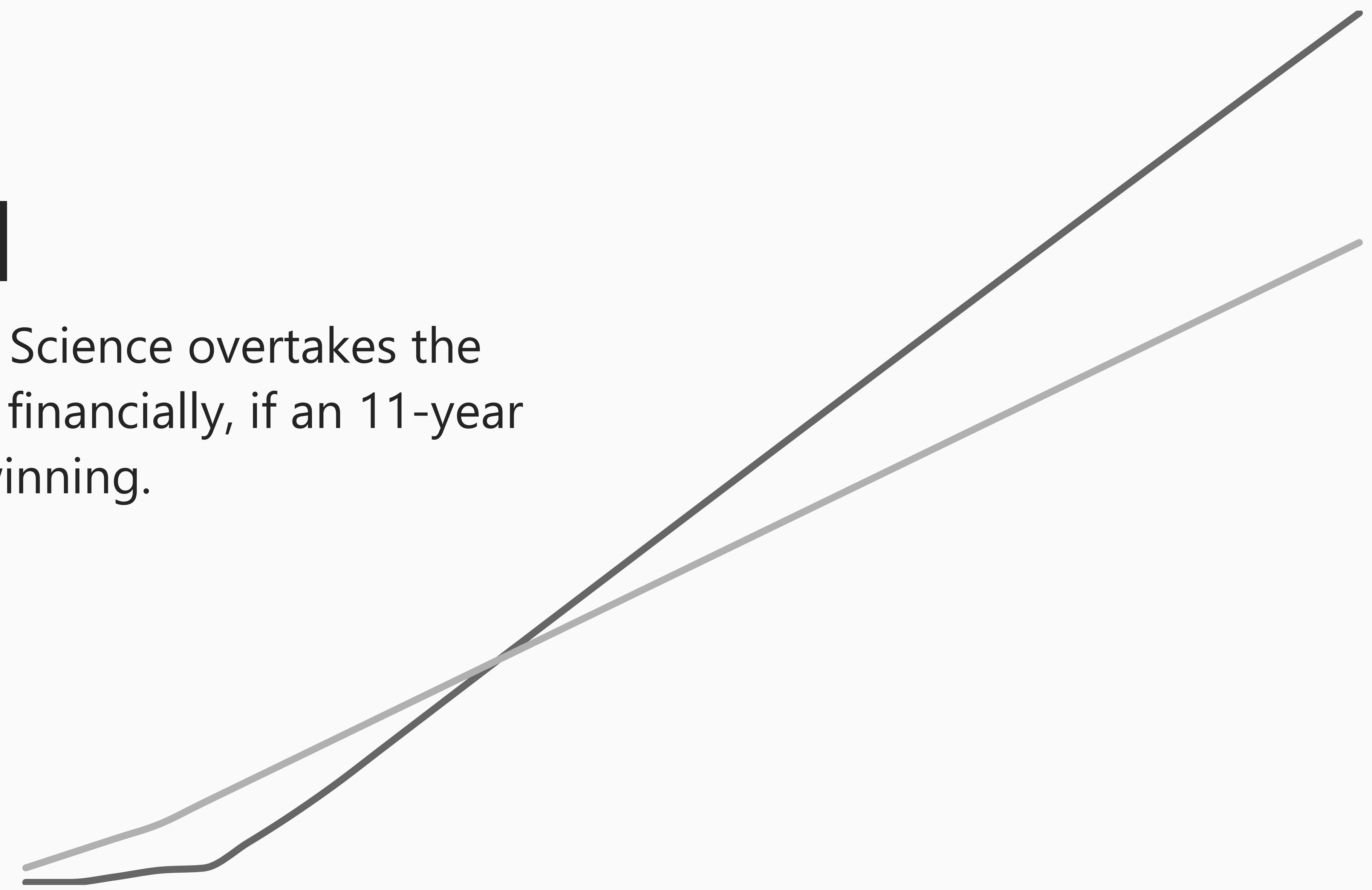
Is a Four-Year Degree Still Worth It?

Comparing Long-Term Financial and Satisfaction Outcomes of College vs. the Electrician Trade

Year 11

when Computer Science overtakes the electrician trade financially, if an 11-year wait counts as winning.

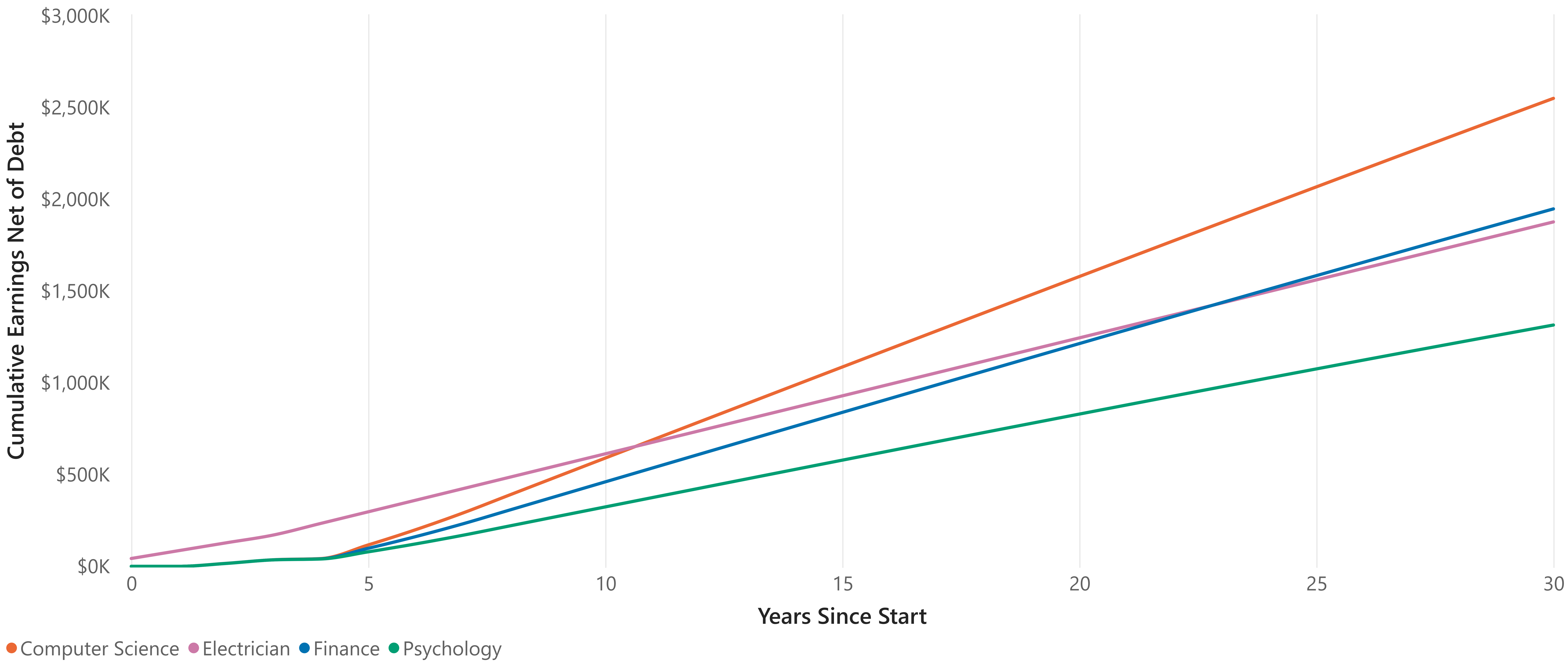


I've spent five years teaching high school math, watching capable students leave for the trades. This summer, working weekends helping my father-in-law renovate his property, I saw why: framing, roofing, running a Bobcat, pouring a concrete sidewalk. It was more problem-solving than I expected walking in. Then I met Fernando, the roofing crew's lead, about my age. When he learned I used to teach math, he laughed, "you probably know a lot of things I can't do." I told him the truth: "yeah, but you know a lot of things I don't." He didn't seem to believe me. I wanted to explore my father-in-law's trade: electrician. The same trade my students keep choosing over college.

Money and satisfaction don't agree on which path is worth it, and neither did that roofer's assumption about which of us knew more.

Computer Science Overtakes Electrician at Year 11. Finance Takes Until Year 23. Psychology Never Catches Up.

Cumulative Earnings Net of Debt, by Career Path



Completion Assumption

Accounts for Dropout Risk

Assumes Everyone Finishes

CS overtakes Electrician in year:

11

Finance overtakes Electrician in year:

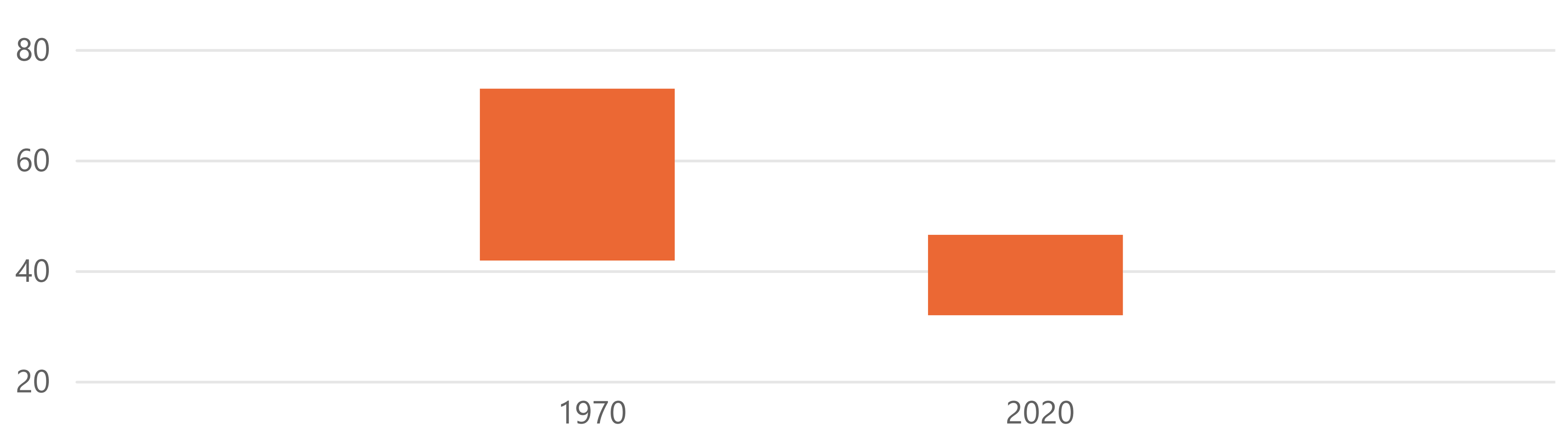
23

Shown here under "Accounts for Dropout Risk" (toggle above to compare). These are cumulative earnings minus debt, assuming flat earnings once each career reaches its ceiling, so catching up financially isn't the end of the story. Whether an 11-year wait is worth it also depends on how satisfying the work stays over time. See Job Satisfaction Trend.

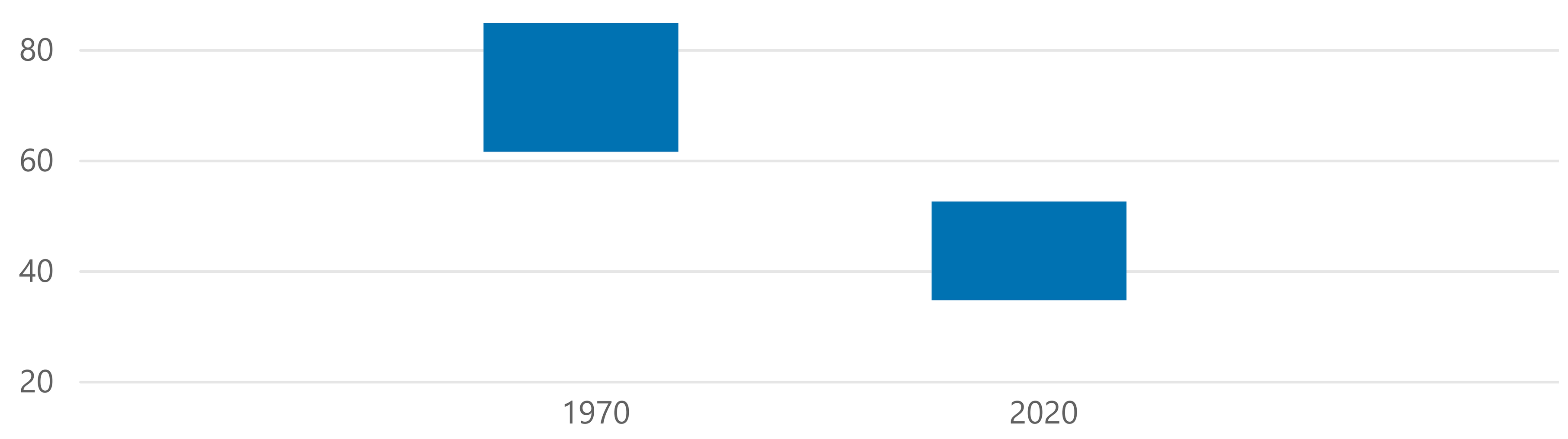
Job Satisfaction Declined Across the Board, But Only Finance's Drop Holds Up Statistically

Job Satisfaction by Decade (% 'Very Satisfied')

CS-related: 1970 (41–73%) vs. 2020 (31–46%). Bands Overlap, Not Confirmed



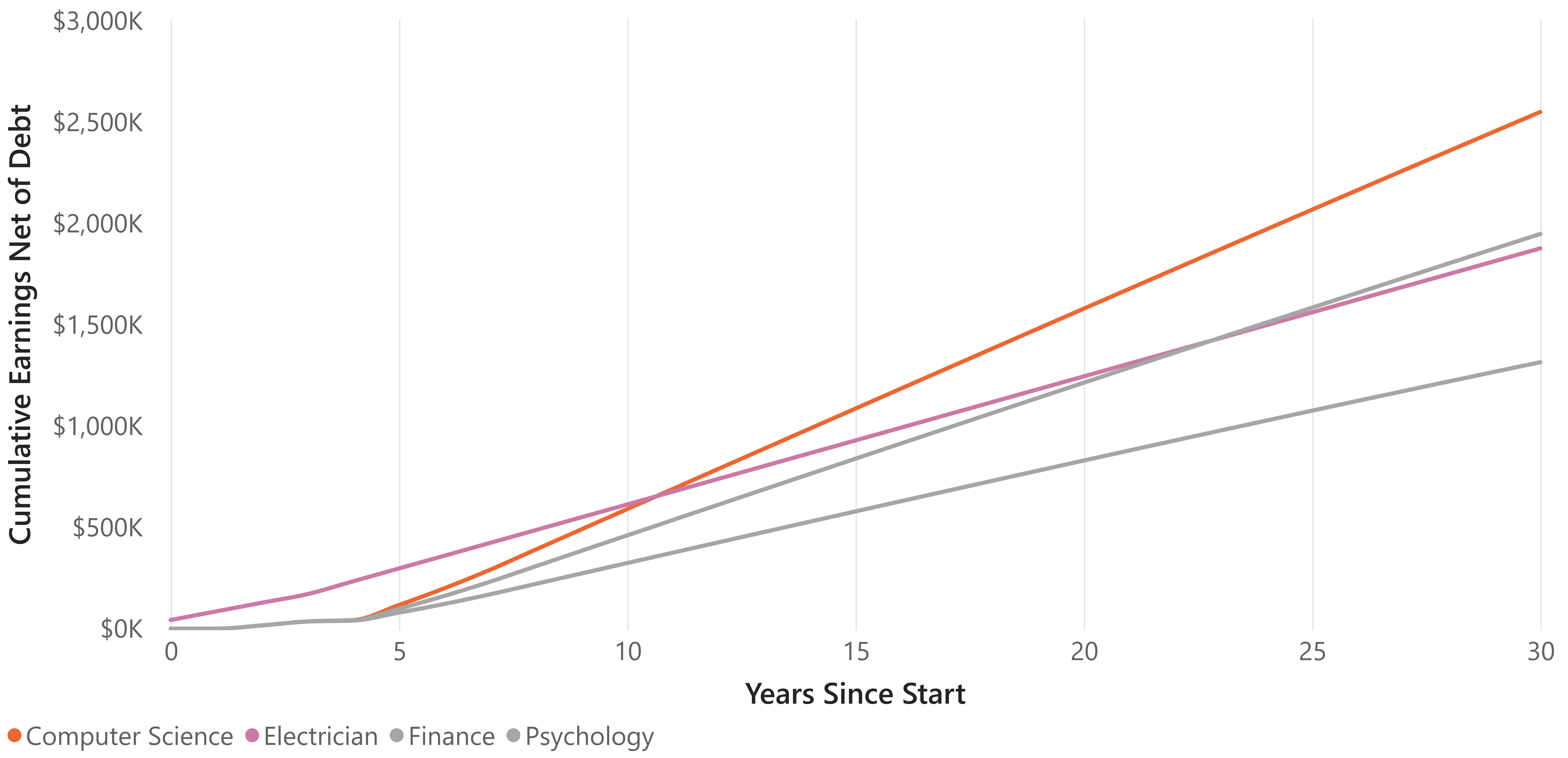
Finance-related: 1970 (61–85%) vs. 2020 (34–52%). No Overlap, Confirmed Decline



Each bar pair compares a field's own 1970 estimate to its 2020 estimate. A gap between the bars means the change is statistically real; touching or overlapping bars means it isn't.

Computer Science Pays Best Long-Term, But Electrician Satisfaction Has Held Flat for 50 Years, at Zero Debt

Cumulative Earnings Net of Debt: Computer Science vs. Electrician



Cost and Payoff Time by Career Path

Career Path	Median Debt	Crossover Year vs Electrician
Electrician	$0	
Computer Science	$10,175	11
Finance	$10,863	23
Psychology	$11,468	

Completion Assumption

Accounts for Dropout Risk

Assumes Everyone Finishes

Electrician satisfaction has stayed roughly flat for five decades (57% in the 1970s, 53% today), and it's the only path with zero debt. CS and Finance both require real debt to reach satisfaction that's comparable to, or lower than, Electrician's. Finance's satisfaction has also seen a statistically confirmed decline over that same span (see Job Satisfaction Trend).

Psychology's crossover-year cell is blank by design, not missing data: under this model's flat-earnings assumption, Psychology never overtakes the electrician path.

In Conclusion: There's No Single "Worth It". It Depends What You're Optimizing For

Early in this project, a roofer about my age, after learning I used to teach high school math, told me, half-joking, "you probably know a lot of things I can't do." I told him the opposite was just as true. My father-in-law has spent his career as an electrician. He genuinely enjoys the work, is the fix-anything person our family counts on, and earns a comfortable wage doing it. He's living proof of the flat satisfaction number in this data: fifty years without a real change, and zero debt, while the "safer" four-year path only wins on paper after an 11-year wait, and only for some fields.

If you're optimizing for:

Zero debt and stability today ➡️ *Electrician:* 50 years of flat satisfaction, zero debt, income now.

Maximum lifetime earnings, if you can wait ➡️ *Computer Science:* overtakes the trade at year 11, if you finish and stay.

One path worth a second look ➡️ *Finance:* slowest payoff (year 23), and the only field with a confirmed satisfaction decline.

A choice for reasons other than pay ➡️ *Psychology:* never overtakes the trade financially here, but pay was never the whole reason to choose it.

Where This Data Comes From, and What It Doesn't Cover

Data Sources:

- College Scorecard (U.S. Department of Education): field-of-study debt, earnings, and completion data.
- Bureau of Labor Statistics, Occupational Employment and Wage Statistics (OEWS): electrician wage data.
- U.S. Department of Labor, Registered Apprenticeship Partners Information Data System (RAPIDS): apprenticeship completion and wage progression.
- General Social Survey (NORC, University of Chicago): job satisfaction by occupation, 1972 to 2024.

Deliberately out of scope (not gaps we missed):

- A second trade beyond electrician (plumbing or HVAC), to test whether these findings hold up past one trade.
- Occupational physical toll (injury/disability rates), a real, commonly-raised argument against the trades that this data doesn't capture. Satisfaction and physical toll are different constructs, and GSS's satisfaction figures may even understate toll: workers forced out by injury exit the surveyed population entirely.